

EDA Summary — Smart MCQ Solver Challenge

Milestone 1: NLP Foundation & Semantic Similarity

Dataset Overview

	Train	Test
Rows	2,000	500
Columns	id, prompt, A, B, C, D, E, answer	id, prompt, A, B, C, D, E
Missing values	0	0

Task: given a prompt and 5 options (A–E), predict the **top 3 most likely correct answers in ranked order**. Evaluated with **MAP@3**.

1. Answer Label Distribution

The correct answer is mildly imbalanced across A–E:

Answer	Count	%
B	490	24.5%
C	459	23.0%
A	369	18.4%
D	358	17.9%
E	324	16.2%

Not severe enough to require resampling, but worth flagging in error analysis — a model with a slight bias toward predicting B/C first would already align with the base rate.

2. Duplicate & Near-Duplicate Prompts

- **454 / 2000** train rows (22.7%) share an **exact duplicate prompt** with at least one other row.
- Of the 212 duplicate-prompt groups, only **158 (74.5%)** have fully consistent options + answer — the rest show minor label/content noise, worth a mention in the report's limitations section.
- After stripping common instruction wrappers (e.g. *"Pick the best possible answer:", "...among the listed options."*) to isolate the **core question**, duplication jumps to **1,227 / 2000** train rows — meaning many prompts are the *same underlying question* reworded with a different instruction template.
- **293 core prompts appear in both train and test**. This is the most actionable finding: a

meaningful chunk of test questions are near-identical to train questions, just phrased differently. This strongly motivates a **similarity-based / retrieval approach** (TF-IDF or embedding lookup against train) as a high-value strategy, and means train/val splits should be **grouped by core prompt** to avoid leakage.

3. Text Length Characteristics

Prompt word count (median 17, range 3–51):

Stat	Value
Mean	18.1
Std	6.8
Min / Max	3 / 51
25% / 50% / 75%	14 / 17 / 22

Option word count (median 22, much wider spread — options range from 1-word answers to full explanatory sentences):

Stat	Value
Mean	26.1
Std	17.2
Min / Max	1 / 81
25% / 50% / 75%	13 / 22 / 37

Train vs test prompt-length distributions overlap closely — no distribution shift between the two sets, so models validated on a train-derived holdout should generalize to test.

4. Option Length vs Correctness

- Correct options average **28.7 words**, incorrect options average **25.7 words** — correct options tend to be slightly longer and more detailed.
 - **"Always pick the longest option" heuristic scores 34.9% accuracy** (vs. 20% random baseline for top-1). This confirms length is a real, exploitable signal — but explains at most a third of cases, so it's a useful weak feature, not a standalone solution.
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5. Instruction Templates ("Wrapper" Phrasing)

Prompts are built from a small set of fixed instruction templates, e.g.:

- *"Pick the best possible answer: ... among the listed options."*

- *"Select the most accurate option: ... carefully."*
- *"Determine the correct option: ... from the following choices."*
- *"Choose the correct answer: ..."*
- *"Identify the correct statement: ... based on the given context."*

These wrappers carry no semantic signal about the answer and should be **stripped before TF-IDF/embedding similarity** — otherwise they add shared noise across all rows (which slightly inflates apparent similarity between unrelated questions).

6. Question-Type Breakdown

The dominant question type is **"What"**, followed by "How", "Which", and "Who". Very few "Why"/"When"/"Where" questions. Most content leans toward **factual/definitional recall** (physics, astronomy, philosophy topics observed in sampled rows) rather than reasoning-heavy questions — relevant context for choosing between retrieval-style vs. reasoning-style modeling approaches.

7. Vocabulary / Content Themes

Word clouds and top n-grams show prompts are dominated by:

- Structural/instructional terms: *given, context, listed, options, following, choices, correct, determine, select*
 - Domain content varies widely (physics, astronomy, philosophy terms appear in sampled examples), suggesting the dataset spans **multiple knowledge domains** rather than one narrow subject — reinforces that a general-purpose semantic similarity approach (TF-IDF/embeddings) is more appropriate than domain-specific rules.
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8. Key Takeaways for Modeling (Next Step: TF-IDF Baseline)

1. **Strip instruction wrappers** before vectorizing — keep only the core question + options.
 2. **Train/test core-prompt overlap (293 rows)** suggests a similarity-lookup against train could give a strong signal for a meaningful subset of test questions.
 3. **Option length is a weak but real signal** (34.9% longest-option accuracy) — consider adding word-count as an auxiliary feature alongside TF-IDF similarity scores.
 4. **No missing data, no distribution shift** between train/test — standard TF-IDF cosine-similarity pipeline (prompt vs. each option) is a clean, well-motivated baseline.
 5. **Group any train/val split by core prompt**, not exact prompt, to avoid near-duplicate leakage inflating validation scores.
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Generated from 01_preprocessing_eda.py and 02_extensive_eda.py — see eda_outputs/ for all supporting plots.